

Data Structures



Lecture 7 Minutes Before Mid

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Quiz problems review

2. Write a function **fun_with_array(arr, i)** that takes a linear array and an integer *i* as input. The function stores the **squares of every *i*th element** of the input array in a new array and returns the new array. See sample input and output for better understanding. **[5]**

arr = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

function call : fun_with_array(arr, 3)

output: [9, 36, 81]

arr = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

function call : fun_with_array(arr, 4)

output: [16, 64]

Quiz problems review

3. Write a function **diagonal_swap(array)** that takes an $n \times n$ array as parameter, **swaps the adjacent pairs of elements in the secondary diagonal** of the array and returns the updated array. [6]

→ You can assume n is an even number.

→ For a 4×4 matrix elements (0,3) and (1,2) are swapped. Similarly, elements (2,1) and (3,0) are swapped. See the sample input and output for better understanding.

→ You cannot create a new array.

→ You cannot use any built-in function except for `len()` and `shape` attribute.

```
arr = [ 1 , 2 , 3 , 4  
        5 , 6 , 7 , 8  
        9 , 10, 11, 12  
        13, 14, 15, 16]
```

function call : `diagonal_swap(array)`

```
output: [ 1 , 2 , 3 , 7  
          5 , 6 , 4 , 8  
          9 , 13, 11, 12  
          10, 14, 15, 16]
```

Mid problems review

To decrypt this message efficiently, you have a task to construct a method called `sum_diff(matrix)` which takes an encrypted matrix as input and returns a decrypted linear array. The process of finding the decrypted linear array is given below:

You have to find out the column-wise summations for each column and store the difference of subsequent column-wise summations in a new linear array.

Sample Input	Sample output	Explanation																			
<table><tr><td>1</td><td>3</td><td>1</td></tr><tr><td>6</td><td>4</td><td>2</td></tr><tr><td>5</td><td>1</td><td>7</td></tr><tr><td>9</td><td>3</td><td>3</td></tr><tr><td>8</td><td>5</td><td>4</td></tr></table>	1	3	1	6	4	2	5	1	7	9	3	3	8	5	4	<table><tr><td>-13</td><td>1</td></tr></table>	-13	1	Sum of 0th column = 29 Sum of 1st column = 16 Sum of 2nd column = 17 Therefore, the size of the resulting array is 2 and the array is: <table><tr><td>16-29 = -13</td><td>17-16 = 1</td></tr></table>	16-29 = -13	17-16 = 1
1	3	1																			
6	4	2																			
5	1	7																			
9	3	3																			
8	5	4																			
-13	1																				
16-29 = -13	17-16 = 1																				

Mid problems review

You are given a non-dummy headed singly linear linked list containing positive integers. Your task is to complete the given method IsSumPossible() which takes the head of the linked list and an integer, n as input. If the integer n can be obtained by summing any two of the elements of the linked list, return True, else return False.

Hint: There might be more than one such pair possible, you don't have to check all such pairs.

[DO NOT USE OTHER DATA STRUCTURE OTHER THAN GIVEN LINKED LIST]

Sample Input	Returned Value	Explanation
list = 1→2→3→4→5 n = 7	True	Sum of the elements (3,4) or (2,5) equals 7
list = 1→2→4→5→6 n = 4	False	There are no two elements that make the sum 4. Note that though there is an element 4 itself in the list, the output will be False.
list = 5 n = 5	False	There is only one element so, the sum of two elements cannot be 5

Mid problems review

Suppose that an intermixed sequence of stack push and pop operations are performed. The pushes push the integers 0 through 9 in order; the pops print out the return value. Which of the following sequences could not occur for push and pop operation? Write all the correct answers.

- A. 1 2 3 4 5 6 9 8 7 0
- B. 0 4 6 5 3 8 1 7 2 9
- C. 1 4 7 9 8 6 5 3 0 2
- D. 2 1 4 3 6 5 8 7 9 0

Mid problems review

You are given a **Stack** of people. They should be paired off as man with woman, woman with man. If a male and a female are consecutive in the stack, they get paired immediately. If there are consecutive females found without any subsequent males, then the spare females should be saved in a **Queue** for future pairing. If there are consecutive males found without any subsequent females, we try to pair a male with the first spare female [if available] from the **Queue**. If no spare female is available, then the male is discarded. The same logic applies to consecutive females without any subsequent males.

Mid problems review

Sample Input	Output	Explanation										
st = <table><tr><td>M_10</td></tr><tr><td>F_20</td></tr><tr><td>F_4</td></tr><tr><td>F_5</td></tr><tr><td>M_3</td></tr><tr><td>M_19</td></tr><tr><td>M_1</td></tr><tr><td>M_7</td></tr><tr><td>F_9</td></tr><tr><td>F_18</td></tr></table>	M_10	F_20	F_4	F_5	M_3	M_19	M_1	M_7	F_9	F_18	10 and 20 are paired together 5 and 3 are paired together 4 and 19 are paired together 7 and 9 are paired together	M_10 and F_20, F_5 and M_3, M_7 and F_9 are sequential in the stack. They are paired up. From F_4, there are consecutive females. So, F_4 is stored in the queue for future pairing. From M_19, there are consecutive males. M_19 is paired with spare F_4. M_1 is discarded since there is no spare female with whom M_1 can be paired.
M_10												
F_20												
F_4												
F_5												
M_3												
M_19												
M_1												
M_7												
F_9												
F_18												

Mid problems review

Complete the function `compress_matrix` that takes a 2D array as a parameter and return a new compressed 2D array. In the given array the number of row and column will always be even. **Compressing a matrix means grouping elements in 2x2 blocks and sums the elements within each block.** Check the sample input output for further clarification.

Hint: Generally the block consists of the (i,j) , $(i+1,j)$, $(i,j+1)$ and $(i+1, j+1)$ elements for 2x2 blocks.

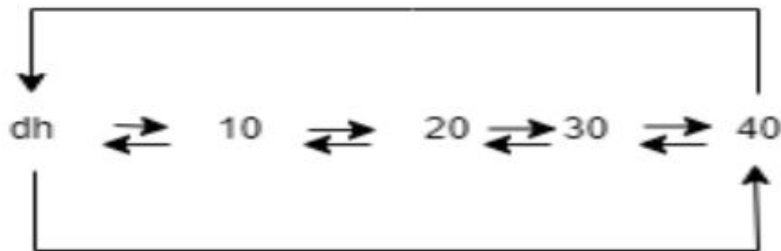
You cannot use any built-in function except `len()` and `range()`. You can use the `np` variable to create an array.

Python Notation	Java Notation
<pre>import numpy as np def compress_matrix (mat): # To Do</pre>	<pre>public int[][] compress_matrix (int[][] mat) { // To Do }</pre>

Sample Input array	All Box (No need to create these arrays)	Returned Array	Explanation
<pre>[[1, 2, 3, 4], [5, 6, 7, 8], [1, 3, 5, 2], [-2, 0, 6, -3]]</pre>	<pre>[[1, 2], [[3, 4], [5, 6]] [[7, 8]] [[1, 3], [[5, 2], [-2, 0]] [[6, -3]]</pre>	<pre>[[14, 22], [2, 10]]</pre>	<pre>[[1+2+5+6, 3+4+7+8], [1+3+-2+0, 5+2+6+-3]]</pre>

Mid problems review

You are given a dummy headed doubly circular linked list and a block of code. The list is like below:



```
for i in range(5):  
    for j in range(i):  
        n1 = dh.next  
        n2 = n1.next  
        n3 = dh.prev  
        dh.next = n2  
        n2.prev = dh  
        n3.next = n1  
        n1.next = dh  
        n1.prev = n3  
        dh.prev = n1
```

Draw the resulting list (The list you will find after the nested loop) for each value **i** in your answer script. Mention the value of **i** and the resulting list, no need to show the intermediate states.

Linked List (Problems)

Deleting all Occurance of a particular element